

Skill-10

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Program:

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tejaswini@teju: ~/OS_Lab/skil × + ∨
GNU nano 8.7.1
// OSSP Skill 10 - pwd, exit, and export
// Demonstrates current directory retrieval, exit handling,
// resource cleanup, export syntax, environment update,
// and inheritance by a child process.

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>
#include <limits.h>
#include <ctype.h>

#ifndef PATH_MAX
#define PATH_MAX 4096
#endif

static void cleanup(void) {
    printf("Cleaning up shell resources...\n");
}

static int valid_name(const char *name) {
    if (!name || (!isalpha((unsigned char)name[0]) && name[0] != '_'))
        return 0;

    for (int i = 1; name[i]; i++)
        if (!isalnum((unsigned char)name[i]) && name[i] != '_')
            return 0;

    return 1;
}
```

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static void do_export(char *text) {
    char *eq = strchr(text, '=');

    if (!eq) {
        printf("Usage: export NAME=value\n");
        return;
    }

    *eq = '\0';
    const char *name = text;
    const char *value = eq + 1;

    if (!valid_name(name)) {
        printf("Invalid environment variable name.\n");
        return;
    }

    if (setenv(name, value, 1) == -1)
        perror("setenv");
    else
        printf("Exported %s=%s\n", name, value);
}

int main(void) {
    char line[256];

    atexit(cleanup);

    printf("=== Skill 10: pwd / export / exit ===\n");

    while (1) {
        char cwd[PATH_MAX];

        printf("envshell> ");
        if (!fgets(line, sizeof(line), stdin))
            break;

        line[strcspn(line, "\n")] = '\0';

        if (strcmp(line, "pwd") == 0) {
            if (getcwd(cwd, sizeof(cwd)))
                printf("%s\n", cwd);
            else
                perror("getcwd");
        } else if (strncmp(line, "export ", 7) == 0) {
            do_export(line + 7);
        } else if (strcmp(line, "show") == 0) {
            const char *demo = getenv("DEMO");
            printf("DEMO=%s\n", demo ? demo : "<undefined>");
        } else if (strcmp(line, "child") == 0) {
            pid_t pid = fork();

            if (pid == -1) {
                perror("fork");
            } else if (pid == 0) {
                execlp("sh", "sh", "-c", "echo Child sees DEMO=$DEMO", NULL);
                perror("execlp");
                _exit(1);
            } else {
                waitpid(pid, NULL, 0);
            }
        } else if (strcmp(line, "exit") == 0) {
            printf("Exit requested.\n");
            break;
        } else if (line[0] != '\0') {
            printf("Unknown command.\n");
        }
    }
}

```

```
    }  
}  
  
return 0;  
}
```

Output:

```
tejaswini@teju:~/OS_Lab/skill$ gcc skill_10.c.c -o skill_10  
tejaswini@teju:~/OS_Lab/skill$ ./skill_10  
== Skill 10: pwd / export / exit ==  
envshell> pwd  
/home/tejaswini/OS_Lab/skill  
envshell> export DEMO=HelloOSSP  
Exported DEMO=HelloOSSP  
envshell> show  
DEMO=HelloOSSP  
envshell> child  
Child sees DEMO=HelloOSSP  
envshell> exit  
Exit requested.  
Cleaning up shell resources...  
tejaswini@teju:~/OS_Lab/skill$ |
```